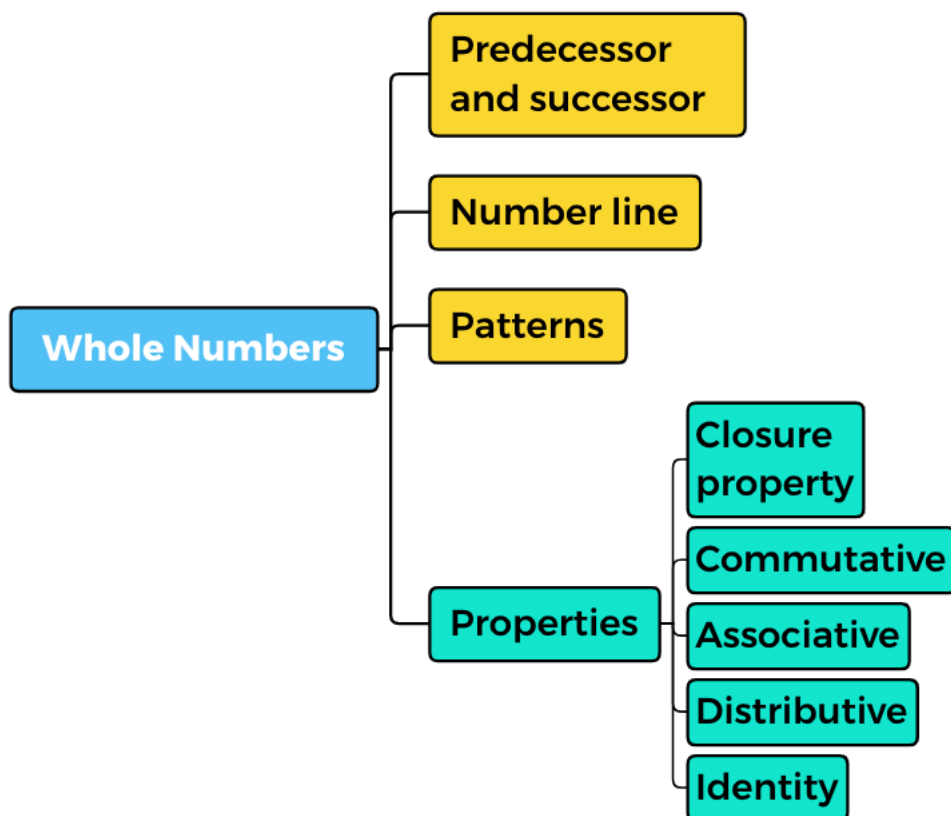


Compendium

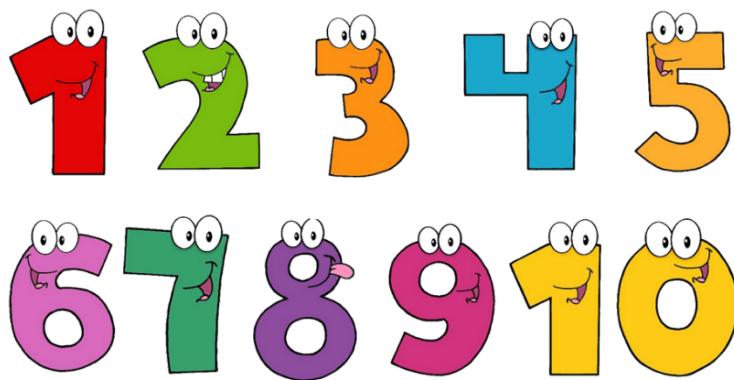
- A. Introduction
- B. Whole numbers
- C. The number line
- D. Properties of whole numbers
- E. Additive inverse of whole numbers
- F. Patterns in whole numbers



A. Introduction

After studying this chapter you will be able to understand idea of whole numbers, their properties, patterns in whole numbers and how to do operations on Whole Number on number line. So let's start learning with basic recap of Natural Numbers.

Natural Numbers: Counting numbers are called natural numbers. It is denoted by N. i.e., $N = 1, 2, 3, 4, 5, \dots$



Expanded form of a natural number

The expanded form of a natural number is written as an addition of the value of each digit.

ONES		
Hundreds	Tens	Ones
3	6	5

This number can also be read as
3 hundred, 6 tens and 5 ones

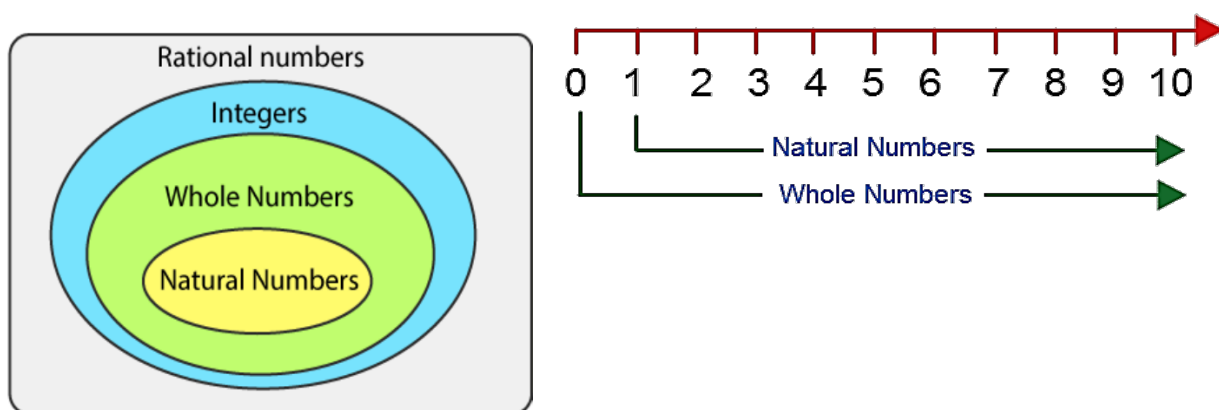
It would look like this in the expanded form:
 $3 \times 100 + 6 \times 10 + 5 \times 1$

CHAPTER 2

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B. Whole numbers

Natural numbers, along with the number 'zero' are called Whole numbers. It is represented as "W", and the set of numbers are {0, 1, 2, 3, 4, 5, 6,}



Successor and Predecessor

The *number immediately before a given number* in counting is known as its predecessor.

Example 1 : 11999

The predecessor of 11999 is $(11999 - 1) = 11998$.

The number *immediately after a given number* in counting is known as successor.

To calculate the successor of any given natural number, simply add 1 to the given number.

Example 2 : 15999

The predecessor of 15999 is $(15999 - 1) = 15998$.



Remember this!

Every number has a successor as we can add 1 to any number.



Recap

What is Successor and Predecessor of a number?

The Successor and Predecessor of a number are explained using examples.

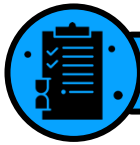
PREDECESSOR	NUMBER	SUCCESSOR
$(7148 - 1)$ 7147	7148	$(7148 + 1)$ 7149
$(8950 - 1)$ 8949	8950	$(8950 + 1)$ 8951
$(7620 - 1)$ 7619	7620	$(7620 + 1)$ 7621
$(12499 - 1)$ 12498	12499	$(12499 + 1)$ 12500

To find the predecessor of a number, subtract 1 from the given number.

To find the successor of a number, add 1 to the given number.

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Checkpoint 1

What is Successor and Predecessor of the mentioned numbers?

Predecessor	Number	Successor
	5665	
	4321	
	8909	
	7000	
	1288	
	3222	

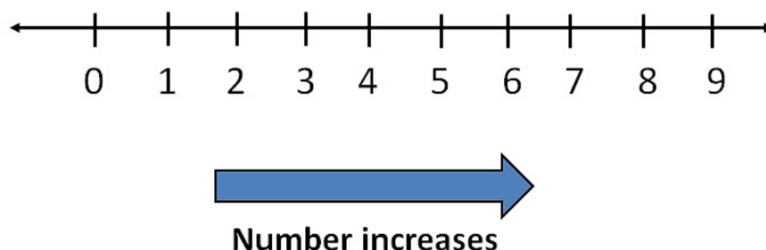


Pro-tip!

The value of nothing is represented by the number **zero**.

C. The number line

In **math**, a **number line** is defined as a straight **line** with **all numbers** placed at equal segments or intervals along its length. A **number** is extended infinitely in any direction but is usually represented horizontally.

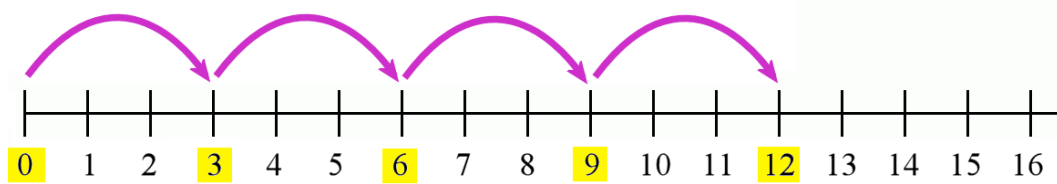


On a number line, when we

- (i) Add a whole number, we move to the right.
- (ii) Subtract a whole number; we move to the left.

Multiplication on the number line

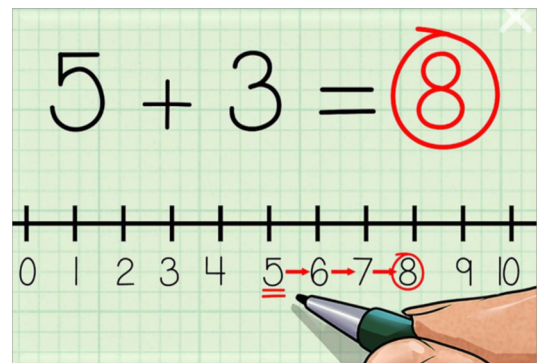
We now see the multiplication of whole numbers on the number line
Let us find 4×3



Start from 0, move 3 units at a time to the right, make such 4 moves.
Where do you reach? You will reach 12. So, we say, $3 \times 4 = 12$.

Example 3 : Solve $5 + 3$ using number line

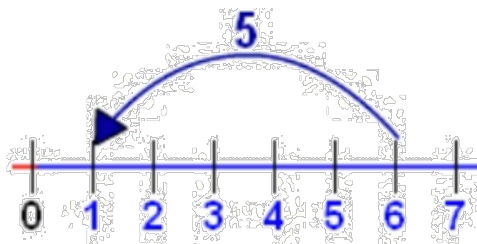
When we add a whole number, we move to the right. Hence;



Example 4 : Solve $(6 - 5)$ using the number line

When we subtract a whole number, we move to the left.

Hence; $6 - 5 = 1$



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Checkpoint 2

Question 1 :

$6 + 5 = 11$	
$9 + 4 =$	
$11 + 5 =$	
$8 + 3 =$	
$2 + 12 =$	
$10 + 7 =$	

Question 2: 20-8



Question 3 : $12 - 5 =$



Question 4 : $5 \times 3 =$

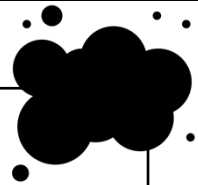


Question 5 : $6 \times 2 =$



Think about this

Zero is a special rational number. It can be written as $0 = \frac{0}{q}$ "where $q \neq 0$



D. Properties of whole numbers

Property 1: Closure Property

Closure property under addition and multiplication states that the sum or product of any two whole number will always be a whole number. i.e. if a and b are any two whole number, $a + b$ and $a \times b$ will also be a whole number.

Example 5 :

$$3 + 4 = 7$$

$$8 \times 5 = 40$$

The results are whole numbers.

Subtraction of whole number :

$$5 - 2 = 3 \text{ (whole number)}$$

$$2 - 5 = -3 \text{ (not a whole number)}$$

So, whole number is not closed under subtraction.



Mental floss!

Whole numbers are not closed under subtraction and division.

Division of whole numbers.

$$7 \div 7 = \frac{7}{7} = 1$$

$$7 \div 5 = \frac{7}{5}$$

Answer is **NOT** always a whole number.

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Closure Property

A set of numbers is said to be 'closed' if the numbers produced under a given operation are also elements of the set.

Tell whether the whole numbers are closed under the given operation. If not, give a counter example.

(1) Addition **Closed**

$$5 + 7 = 12$$

(2) Multiplication **closed**

$$6 \times 0 = 0$$

(3) Subtraction **Not Closed**

$$5 - 7 = -2$$

(4) Division **Not Closed**

$$2 \div 8 = 0.25$$

Property 2: Commutative Property

The word "**commutative**" is derived from "commute" or "to move around". So the **Commutative Property** refers to moving numbers around. For addition, the rule is " $p + q = q + p$ "; in numbers, this means $5 + 3 = 3 + 5$. For multiplication, the rule is " **$xy = yx$** "; in numbers, this means $5 \times 3 = 3 \times 5$.

When we add:

$$p + q = q + p$$

Commutative Property

$$a + b = b + a$$

$$xy = yx$$

$$8 \times 2 = 16$$

1	3	5	7	9	11	13	15
2	4	6	8	10	12	14	16

$$2 \times 8 = 16$$

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16

Commutative Property	
Addition You can add in any order. $a+b=b+a$ $3+5=5+3$	Multiplication You can multiply in any order. $a \times b = b \times a$ $2 \times 6 = 6 \times 2$

Thus, we conclude that addition & multiplication is commutative for Whole Numbers.

Subtraction of any two whole numbers is not commutative.

Let a and b be two integers numbers,

Then $a - b \neq b - a$.

Take $a = 5$ and $b = 3$,

$$5 - 3 = 2 \neq 3 - 5 = -2.$$

$a - b \neq b - a$

Property 3: Associative Property

According to the **associative property of addition**, the sum of three or more numbers remains the same regardless the grouping of numbers.

For example, the sum does NOT change irrespective of how the addends are grouped in

$$(75 + 81) + 34 = 75 + (81 + 34)$$

$$= 156 + 34 = 199.$$

(i.e. which we calculate first) ... when we **add & multiply**:

$(a + b) + c = a + (b + c)$

&

$(a \times b) \times c = a \times (b \times c)$

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$$(3 + 4) + 5 = 3 + (4 + 5)$$
$$7 + 5 = 3 + 9$$
$$12 = 12$$

$$(5 \times 7) \times 3 = 5 \times (7 \times 3)$$
$$35 \times 3 = 5 \times 21$$
$$105 = 105$$

An associative property does not hold true for the subtraction & division of whole numbers. This concludes that we cannot group any two whole numbers and subtract/divide them first. Order of subtraction/division is a very important factor. If 'p', 'q', and 'r' are the three numbers then,

$$(p - q) - r \neq p - (q - r)$$

Consider the case when $p = 8$, $q = 5$ and $r = 2$,

$$8 - (5 - 2) = 5 \neq (8 - 5) - 2 = 1$$

$$100 \div 20 \div 5 = 100 \div 20 \div 5$$
$$(100 \div 20) \div 5 \neq 100 \div (20 \div 5)$$
$$(5) \div 5 \neq 100 \div (4)$$
$$1 \neq 25$$

Property 4: Distributive Property

The sum of the product of a whole number with two other whole numbers is equal to the product of the whole number with the sum of the two other whole numbers. This called the **distributive property of multiplication over addition**.

The distributive property allows you to multiply a sum by multiplying each addend separately, then add the products.

To understand more easily, look at the examples below.

Consider the first example, the distributive property lets you "distribute" the 5 to both the 'x' and the '2'.

$$5(x + 2) = 5 \cdot x + 5 \cdot 2$$

$$(5x)(3x + 6) = 5x \cdot 3x + 5x \cdot 6$$

$$(5)(3x^2 + 2x + 6) = 5 \cdot 3x^2 + 5 \cdot 2x + 5 \cdot 6$$

If, for some reason, you are having trouble accepting the distributive property, look at the examples below.

Below 2 examples show that you can apply this property or formula to numbers as well as expressions.

Normal Way	Distributive Approach
$(5)(\underline{8}) = 40$	$5(\underline{6+2}) = 5 \cdot 6 + 5 \cdot 2$ $= 30 + 10$ $= 40$
$(4)(\underline{12}) = 48$	$(4)(\underline{7+2+3}) = 4 \cdot 7 + 4 \cdot 2 + 4 \cdot 3$ $= 28 + 8 + 12$ $= 48$

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Property 5: Multiplicative identity of whole number

The "Multiplicative Identity" is **1**, because if we multiply a number by 1, it remains unchanged:

$$a \times 1 = 1 \times a = a$$

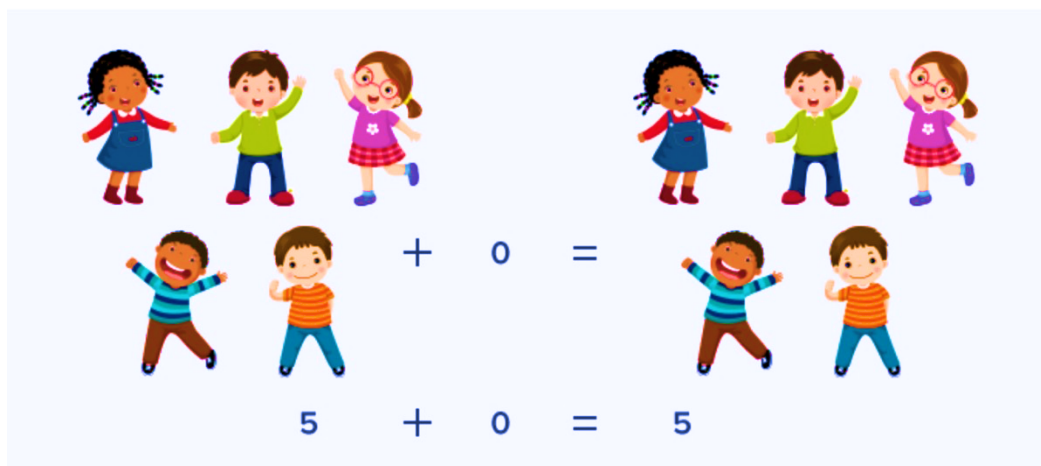
Remember: One (1) preserves the identity of every number during multiplication.

Property 6: Additive identity of whole number

When we add zero to any whole number, we get the same whole number.

Zero is an additive identity for whole number. Observe the following and fill in the blanks: Let a be a whole number

$$a + 0 = a$$

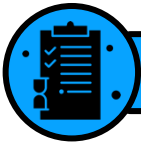


E. Additive inverse of whole number

The additive inverse of any number a is $-a$.

In other words, the additive inverse of a whole number **a** is what you add to **a** to get **Zero (0)**.

$$a + (-a) = 0$$



Checkpoint 3

- Which of the following is not a natural number?
(A) 5 (B) -7 (C) 18 (D) 63
- Which of the following options is the same as $45(12 - 7) = ?$
(A) $(45-7) \times (45-13)$ (B) $57-7$
(C) $(45 \times 12) \times 45 \times 7$ (D) $(45 \times 12) - (45 \times 7)$
- $654 \times (182 - 82) = ?$
(A) 75,400 (B) 65,400 (C) 65,4000 (D) none
- Which statement is NOT correct?
(A) The Commutative Property is true for multiplication
(B) The Commutative property is true for addition
(C) The Commutative property is true for subtraction.
(D) The Commutative property is not true for division.
- A number is divisible by 8 if the number formed by the last three digits is divisible by _____.
(A) 2 (B) 4 (C) 6 (D) 8
- Which of these are two consecutive numbers after 70009?
(A) 700100, 70011 (B) 70010, 700011
(C) 70010, 70011 (D) 7010, 7011
- $56(4 + 2) = (56 \times 4) + (56 \times 2)$ is an example of the _____.
(A) closure property of whole number
(B) commutative property of whole number
(C) Associative property of whole number
(D) Distributive property of whole number
- Which is an example of Commutative Property?
(A) $21 \times 1 = 21$ (B) $5 \times (6 + 7) = 5 \times 6 + 5 \times 7$

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(C) $12+32=32+12$

(D) $12+0=12$

9. What least number should be subtracted from 1,222 to get a number exactly divisible by 42?

(A) 12

(B) 3

(C) 1

(D) 4

10. Which of the below options is an example of Associative property of addition?

(A) $a+(b+c)=ab+ac$

(B) $a+(b+c)=b(a+c)$

(C) $a+(b+c)=a(+b)+c$

(D) $(a+b)\times c$

F. Patterns in whole number

A pattern is a logical sequence of numbers or pictures.

We can arrange these numbers in elementary shapes consisting of points or dots.

1. Every number can thus be arranged as a line.

Number 1 is shown as



Number 2 is shown as



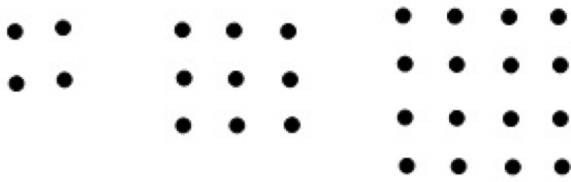
Number 3 is shown as



2. Some numbers like 3, 6 and 10 can be arranged as a triangle.



3. Some numbers like 9, 16 and 25 can be arranged as a square



4. Some numbers like 6, 8 and 10 can be arranged as a rectangle.



Identify the geometrical shapes formed by joining the given number of points :

Number of points	Lines	Triangles	Squares	Rectangles
2	YES	NO	NO	NO
3	YES	YES	NO	NO
4	YES	NO	YES	YES
5	YES	NO	NO	NO
6	YES	YES	NO	YES
7	YES	NO	NO	NO
8	YES	NO	NO	YES
9	YES	NO	YES	NO
10	YES	YES	NO	YES

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Patterns observations:

Example 6 : Multiply 3 by 9, 99, 999, 9999. Study the pattern.

Solution:

$$4 \times 9 = 4 \times (10 - 1) = 36$$

$$4 \times 99 = 4 \times (100 - 1) = 396$$

$$4 \times 999 = 4 \times (1000 - 1) = 3996$$

$$4 \times 9999 = 4 \times (10000 - 1) = 39996$$

Example 7 : Observe the following pattern.

$$64 \times 5 = 64 \times 10/2 = 32 \times 10 = 320 \times 1$$

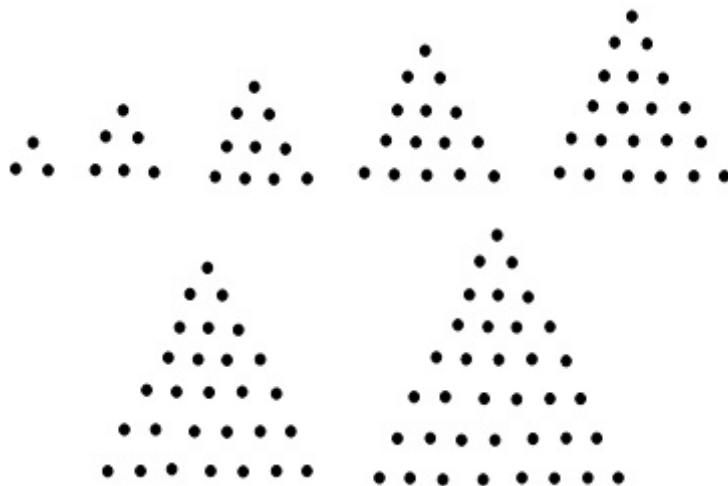
$$64 \times 10 = 64 \times 20/2 = 32 \times 20 = 320 \times 2$$

$$64 \times 15 = 64 \times 30/2 = 32 \times 30 = 320 \times 3$$

$$64 \times 20 = 64 \times 40/2 = 32 \times 40 = 320 \times 4$$

Example 8 : Write the first seven numbers that can be arranged as triangles, e.g. 3, 6, ...

Solution:



Therefore , first 7 numbers which can be arranged as triangles are 3, 6, 10, 15, 21, 28, 36.

Example 9 : Study the pattern:

$$1 \times 8 + 1 = 9$$

$$12 \times 8 + 2 = 98$$

$$123 \times 8 + 3 = 987$$

$$1234 \times 8 + 4 = 9876$$

$$12345 \times 8 + 5 = 98765$$

Example 10: Write the next two steps. Can you say how the pattern works?

(Hint: $12345 = 11111 + 1111 + 111 + 11 + 1$).

Solution:

$$123456 \times 8 + 6 = 987648 + 6 = 987654$$

$$1234567 \times 8 + 7 = 9876536 + 7 = 9876543$$

Yes, the pattern works.

$$\mathbf{123456} = 111111 + 11111 + 1111 + 111 + 11 + 1$$

$$\begin{aligned} \mathbf{123456 \times 8} &= (111111 + 11111 + 1111 + 111 + 11 + 1) \times 8 \\ &= 111111 \times 8 + 11111 \times 8 + 1111 \times 8 + 111 \times 8 + 11 \times 8 + 1 \times 8 \\ &= 888888 + 88888 + 8888 + 888 + 88 + 8 = 987648 \end{aligned}$$

$$\mathbf{123456 \times 8 + 6} = 987648 + 6 = 987654$$

Number Patterns Observation

Observation of number patterns can lead to simple processes and make the calculations simpler.

Let's understand the following examples which helps us in addition and subtraction with numbers like 9, 99, 999, etc.

- $165 + 9 = 165 + 10 - 1 = 175 - 1 = 174$
- $165 - 9 = 165 - 10 + 1 = 155 + 1 = 156$
- $165 + 99 = 165 + 100 - 1 = 265 - 1 = 264$
- $165 - 99 = 165 - 100 + 1 = 65 + 1 = 66$

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Let's understand another pattern which makes multiplication with 9, 99, 999, and so on simpler:

- $52 \times 9 = 52 \times (10 - 1) = 62 \times 9 = 468$
- $52 \times 99 = 52 \times (100 - 1) = 62 \times 99 = 5148$
- $52 \times 999 = 52 \times (1000 - 1) = 62 \times 999 = 51948$

Let's understand the following pattern which simplifies multiplication with numbers like 5, 25, 125, etc.:

- $58 \times 5 = 58 \times 10/2 = 580/2 = 290$
- $58 \times 25 = 58 \times 100/4 = 5800/4 = 1450$
- $58 \times 125 = 58 \times 1000/8 = 58000/8 = 7250$

A number of patterns of similar kind can be observed in the whole number system which simplifies calculations a lot.



Checkpoint 4

Question 1

Write the least natural and least whole number.

Question 2

Match the column

Closure Property	If p and q are any two whole numbers, then $p + q = q + p$ and $p \times q = q \times p$.
Commutative Property	If p and q are any two whole numbers, then $p + q, q \times p$ are also whole numbers.
Associative Property	If p, q and r are any two whole numbers, then $p(q + r) = p \times q + p \times r$.

Distributive property	If p, q and r are any two whole numbers, then $(p + q) + r = p + (q + r)$ and $(p \times q) \times r = p \times (q \times r)$.
Additive Identity	If a is any whole number, then $p + 0 = p = 0 + p$.
Multiplicative Identity	If a is any whole number, then $p \times 0 = 0 = 0 \times p$.
Multiplication by zero	If p is any whole number, then $p \times 1 = p = 1 \times p$
Division by zero	If p is any whole number, then $p \div 0$ is not defined

Question 3

Match the column

$191 + 13 = 13 + 191$	Associative Property of Multiplication.
$90 + 0 = 90$	Distributive Property of Multiplication over Addition.
$(78 + 1) + 11 = 78 + (1 + 11)$	Commutative Property of Multiplication
$(121 \times 4) \times 80 = 121 \times (4 \times 80)$	Distributive Property of Multiplication over Subtraction.
$12 \times (10 + 85) = 12 \times 10 + 12 \times 85$	Associative Property of Addition
$71 \times (11 - 3) = 71 \times 11 - 71 \times 3$	Additive Identity
$9 \times 45 = 45 \times 9$	Commutative Property of Addition.

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Question 4

Fill up the gaps

- (a) ____ $\times$ 12 = 12 $\times$ 18
 - (b) Whole numbers are closed under ____ and ____ operation.
 - (c) Division by ____ is not defined.
 - (d) ____ is the identity for multiplication.
 - (e) If ____ is added to a number, the sum will remain the same.
- Hence ____ is called the ____ in the whole numbers.

Question 5

Find the number of whole numbers from 12 and 86

Question 6

Find the product of the following numbers using Distributive property

- (a) 168×102
- (b) $625 \times 279 - 625 \times 79$

Question 7

Find the predecessor and successor of each of the given whole numbers:

- (i) 999
- (ii) 21998 (iii) 4001
- (iv) 500014 (v) 11111

Question 8

Seeta got 99 marks in Math, 69 marks in English, and 91 in Hindi. Another student Rima got 92 marks in Maths, 33 in English and 84 in Hindi. What are their total individual marks?

Question 9

Rakesh ordered 10 cartons of chocolates to distribute among the class. Each carton contains 15 boxes, and each box has 12 chocolates. How many chocolates did Rakesh order altogether?

Question 10

Mahesh lives in a hostel which charges Rs 55 for Dinner and 40 for Lunch. Find the total money he has to pay for ten days.

Question 11

Out of 185000 tablets of Vitamin A, 18534 are distributed among the students in a city. Find the number of the remaining vitamin tablets.

Question 12**Fill in the blanks**

(a) $15 \times 38 = 15 \times 18 + 15 \times \underline{\hspace{2cm}}$

(b) $786 \times 8 + 786 \times 2 = \underline{\hspace{2cm}}$

(c) $1001 \times 2002 = 1001 \times (2000 + \underline{\hspace{2cm}})$

(d) predecessor of 1 lakh is $\underline{\hspace{2cm}}$

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Bird's-eye view

- Counting numbers are called **Natural Numbers**. It is denoted by N .
 $\{1, 2, 3, 4, 5, 6, 7, 8, 9, \dots\}$
- Natural numbers together with the number zero are called Whole numbers. It is represented as "W", and the set of numbers are $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, \dots\}$
- In **math**, a **number line** is defined as a straight **line** with **all numbers** placed at equal segments or intervals along its length. A **number** is extended infinitely in any direction but is usually represented horizontally
- Three natural numbers after 11999 = 12000, $12000 + 1 = 12001$, $12001 + 1 = 12002$ etc.
- Predecessor of 103 = $103 - 1 = 102$
- Successor of 101 = $101 + 1 = 102$
- The number 205 lies to left or right of 212? Ans: 205 is less than 212; thus, it lies to the left.
- Closure property** under addition and multiplication states that the sum or product of any two whole number will always be a whole number. i.e. if a and b are any two whole number, $a + b$ and $a \times b$ will also be a whole number.
- The sum of two whole numbers will always be the same, and it does not depend on the order in which they are added. This property is known as the **commutative property of addition**.
- The product of two whole numbers is the same, no matter in which order they are multiplied. This property is known as the **commutative property of multiplication**.

- Subtraction and division are not **commutative in whole numbers**.
- A whole number added to 0 remains unchanged. Thus, 0 is called the **additive identity** in whole numbers.
- A whole number multiplied by 1 remains unchanged. Thus, 1 is known as the **multiplicative identity in whole numbers**.
- While adding whole numbers, we can group the numbers in any order. This property is called the **associative property of addition**.
- While multiplying whole numbers, we can group the numbers in any order. This is called the **associative property of multiplication**.
- The additive inverse of any number a is $-a$.
- In other words, the additive inverse of a whole number a is what you add to a to get **Zero (0)**. A whole number added to 0 remains unchanged. Thus, 0 is called the **additive identity** in whole numbers.
- A whole number multiplied by 1 remains unchanged. Thus, 1 is known as the **multiplicative identity in whole numbers**.
- While adding whole numbers, we can group the numbers in any order. This property is called the **associative property of addition**.
- While multiplying whole numbers, we can group the numbers in any order. This is called the **associative property of multiplication**.



Solved examples

Example 1: Add 342, 45 and 55.

Solution In this case, it is simpler to add 45 and 55 first, to get 100. $342 + 45 + 55 = 342 + (45 + 55) = 342 + 100 = \mathbf{442}$

Example 2: Find the sum of:

(i) 753, 807 and 947

(ii) $1456 + 3457 + 4544$

Solution (i) $753 + 807 + 947 = 753 + 947 + 807 = (753 + 947) + 807 = 1700 + 807 = \mathbf{2507}$

(ii) $1456 + 3457 + 4544 = \underline{1456 + 4544} + 3457 = (1456 + 4544) + 3457 = 6000 + 3457 = \mathbf{9457}$

Example 3: Find: $(23 \times 25) \times 4$

Solution $(23 \times 25) \times 4 = 23 \times (25 \times 4) = 23 \times 100 = \mathbf{2300}$

Example 4: Find: $(1234 \times 25) \times 40 \times 6$

Solution: $(1234 \times 25) \times 40 \times 6 = 1234 \times (25 \times 40) \times 6 = 1234 \times 1000 \times 6 = 12,34,000 \times 6 = \mathbf{74,04,000}$

Example 5: Multiply the following using distributive property.

(i) $(28 \times 36) + (28 \times 64)$

(ii) $(39 \times 26) - (39 \times 16)$

(iii) 536×105

(iv) 2485×148

Solution:

(i) $(28 \times 36) + (28 \times 64) = 28 \times (36 + 64) = 28 \times 100 = 2800$

(ii) $(39 \times 26) - (39 \times 16) = 39 \times (26 - 16) = 39 \times 10 = 390$

(iii) $536 \times 105 = 536 \times (100 + 5) = (536 \times 100) + (536 \times 5) = 53,600 + 2680 = 56,280$

(iv) 2485×148

		2	4	8	5	
		×	1	4	8	
1	9	8	8	0	→	2485×8
9	9	4	0	0	→	2485×40
2	4	8	5	0	0	→ 2485×100
						$2485 \times (8 + 40 + 100) =$
3,	6	7,	7	8	0	→ 2485×148

Example 6: Divide 94,335 by 82 and check your answer by using the division rule.

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Solution:

$$\begin{array}{r} 1150 \\ 82 \overline{) 94335} \\ \underline{82} \\ 123 \\ \underline{82} \\ 413 \\ \underline{410} \\ 35 \end{array}$$

Quotient: **1150**

$$94,335 \div 82 = 1150, \text{ remainder } 35$$

$$\text{Using the division rule: } 82 \times 1150 + 35 = 94,300 + 35 = 94,335$$

Example 7: Identify the numbers below 10 as prime and composite. Identify the triangular numbers.

Solution: The number 1 is a unique number.

The numbers 2, 3, 5 and 7 are prime numbers as they can be represented only as a single rectangular array.

The numbers 4, 6, 8, 9 can be represented as a square or a rectangle at least in two ways. They are therefore composite numbers.

The numbers 1, 3 and 6 are triangular numbers.

3 is a prime number while 6 is a composite number.

• • • • • • • • • • • • • • • • •

1×2 1×3 1×5 1×7

Numbers	Triangular	Composite	
3	3 	No	
4	No	 1×4	 $2 \times 2,$
6	6 	 1×6	 2×3
8	No	 1×8	 2×4
9	No	 1×9	

9. The predecessor of which of the two-digit numbers has a single-digit.

- (a) 20 (b) 9 (c) 10 (d) 11

10. Which natural number has no predecessor

- (a) 0 (b) 1 (c) 10 (d) 100

Short answer type [Each carries variable mark(s)]

11. Write the largest natural and least whole number.

12. Name the property from below

- (a) $19 + 62 = 62 + 19$ (b) $30 + 0 = 30$
(c) $(10 + 3) + 15 = 10 + (3 + 15)$ (d) $(62 \times 4) \times 30 = 62 \times (4 \times 30)$
(e) $40 \times 30 = 30 \times 40$ (f) $86 - 42 = 42 - 86$
(g) $7 \times (8 - 3) = 7 \times 8 - 7 \times 3$ (h) $22 \times (50 + 16) = 22 \times 50 + 22 \times 16$

13. Fill in the blanks (1 x 5)

- (a) $\underline{\hspace{1cm}} \times 73 = 73 \times 24$
(b) Whole numbers are closed under $\underline{\hspace{1cm}}$ and $\underline{\hspace{1cm}}$ operation.
(c) Division by $\underline{\hspace{1cm}}$ is not defined.
(d) $\underline{\hspace{1cm}}$ is the identity for multiplication.
(e) If $\underline{\hspace{1cm}}$ is added to any number, the sum will always remain the same. Hence $\underline{\hspace{1cm}}$ is called the $\underline{\hspace{1cm}}$ in the whole number system.

14. How many whole numbers are there between 52 and 73, Write some examples ?

15. Find the product using Distributive property.

- (a) 838×103 (b) $91625 \times 179 - 91625 \times 79$

16. Find the product by suitable rearrangement: (2 marks)

- (a) $8 \times 391 \times 125$ (b) $2 \times 1234 \times 50$
(c) $87 + 64 + 36$ (d) $713 + 87 + 200$

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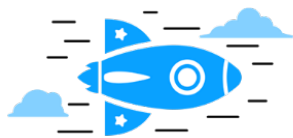
17. Sheldon got 49 marks in Math, 39 marks in English, and 51 in Hindi. Johnny got 62 marks in Maths, 32 in English and 54 in Hindi. What are their total individual marks?
18. The total number of students in each class of school is 30. The fees paid by an individual student is \$ 812 per month. If there are 30 classes in the school, what is the total fee collection in a month?
19. In a flower bouquet, there are 7 roses and 8 gladioli. In 9 bouquets, how many total flowers are there? Write a mathematical statement for this.
20. Jaya eats form a hotel which charges \$ 55 for lunch and \$ 40 for dinner. Find the money she has to pay for seven days.
21. How many times does the digit 5 occur in tens place between natural numbers 100 and 1000?
22. How many times does the digit 8 occur in the first 100 natural numbers?
23. Find the sum by suitable rearrangement.
- (a) $534 + 100 + 466$ (b) $365 + 700 + 635$
24. Add each of the following numbers and check by reversing the order to the addends.
- (a) $3025 + 1825$ (b) $2094 + 2942$
25. Find the product of the following numbers using properties of whole numbers.
- (a) 155×601 (b) 1800×499 (c) 71×380 .
26. Find the value of the following using properties of whole numbers.
- (a) $4,380 \times 85 - 4,380 \times 75$ (b) $270 \times 8 + 270 \times 9 + 270$
- (c) $45 \times 20 \times 5$
27. Find the quotient and remainder in each of the following divisions. Check the answer by division algorithm.

(a) $4,567 \div 234$

(b) $3,645 \div 15$

(c) $7,654 \div 19$

28. The product of two numbers is 3,75,200. If one of the numbers is 160. Find the other?
29. On dividing 4,56,789 by a certain number the quotient is 1,952 and a remainder 21 is left. Find the number.
30. Divide the largest 4 – digit number by the largest 2 digit number and write down the quotient and the remainder.



ADVANCE LEVEL

Subjective type questions (each carries four marks)

- Calculate using suitable rearrangements:
 - $31 + 32 + 33 + 34 + 35 + 65 + 66 + 67 + 68 + 69$
 - $1 + 2 + 3 + 4 + 996 + 997 + 998 + 99$
 - $14 + 16 + 18 + 20 + 22 + 80 + 82 + 84 + 86 + 88$
- What is the difference between the largest 5 digit number and the least 6 digit number?
- The digits of 8 and 9 of the number 38490 are interchanged. Find the difference between the new number and the original number.
- Determine the products by suitable rearrangement:
 - $16 \times 125 \times 64 \times 25$
 - $250 \times 40 \times 50 \times 16$
 - $37256 \times 125 \times 9 \times 8$
- Determine the product of:
 - The largest 4-digit number and the smallest 3-digit number

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- (ii) Smallest number of 2-digits and the greatest number of 5-digits.
6. A dealer purchased 120 LCD television sets. If the cost one set is Rs. 25000, determine the total cost of all sets.
7. Find the value of each of the following using properties:
- (i) $493 \times 9 + 493 \times 2$ (ii) $24579 \times 93 + 7 \times 24579$
- (ii) $1568 \times 184 - 1568 \times 84$ (iv) $5625 \times 1625 - 5625 \times 625$
8. The product of two whole numbers is zero. What do you conclude?
9. Determine the products by suitable rearrangement:
- (i) $2 \times 1497 \times 50$ (ii) $4 \times 358 \times 25$ (iii) $625 \times 20 \times 8 \times 50$
10. Find the product 8739×102 using the distributive property.
11. Write in expanded form :
- (a) 74836 (b) 574021 (c) 8907010
12. A person has Rs 1000000 with him. He purchases a colour T.V. for Rs 16580, a motorcycle for Rs 45890 and a flat for Rs 870000. How much money is now left with him?
13. Out of 185000 tablets of Vitamin A, 18334 are distributed among the children in a district. Find the number of vitamin tablets remaining.
14. Carol had Rs 620000. She gave Rs 97500 to Jyoti, Rs 126380 to Jawed and Rs 350000 to Shailja. How much money was left with her?
15. Find the difference between the highest number of seven digits and the least number of eight digits.
16. A cell phone number consists of a total of ten digits. The first four digits are 8, 5, 9 and 7. The last three digits are 3, 8 and 5. The remaining digits are unique and make the cell phone number, the greatest possible number. What are these digits?

17. A cell phone consists of ten digits. First four digits are 9,8,7 and 9. Make the smallest cell phone number by using only one digit twice from 8, 3, 5, 6, 0.
18. In a particular five-digit number, digit at ten's place is 8, digit at unit's place is one-fourth of ten's place digit, digit at hundreds place is 0, digit at thousand's place is 4 times of the digit at unit's place and ten thousand's place digit is equal to the digit at ten's place. Write the number.
19. Find the sum of the largest and the smallest six-digit numbers formed by the digits 2, 0, 4, 7, 6, 5 using each digit only once.
20. An industry has a container filled with 35874 litres of cold drink. In how many bottles of 300 ml capacity each can it be filled?
21. The population of a city is 450772. In a survey, it was being reported that one out of every 7 persons is illiterate. Find the number of illiterate persons in the city.
22. Find the sum of the four numbers as mentioned below:
 - (a) Successor of 32
 - (b) Predecessor of 49
 - (c) The predecessor of the predecessor of 56
 - (d) The successor of the successor of 67
23. A loading truck can carry 485 boxes of biscuits weighing 15kg each, whereas a tempo can carry 520 boxes each of the same weight. Find the total weight of biscuits that can be carried together by both the vehicles.
24. In the marriage of her daughter, Lata spent Rs 218766 on food and decoration, Rs. 125322 on jewellery, Rs 98234 on furniture and Rs 26780 on miscellaneous items. Find the total amount she spent at the wedding.

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25. A box contains 6 strips having 15 capsules of 500mg medicine in each capsule. Calculate the total weight in grams of medicine in 32 such boxes.



Solutions

Check point 1

Predecessor	Number	Successor
5664	5665	5666
4320	4321	4322
8908	8909	8910
6999	7000	7001
1287	1288	1289
3221	3222	3223

Check point 2

- 13 16 11 14 17
- 12
- 7
- 15
- 12

Check point 3

1	2	3	4	5	6
b	d	b	c	d	c
7	8	9	10		
d	c	d	c		

Check point 4

1. 1 & 0
- 2.

Closure Property	If p and q are any two whole numbers, then $p + q$, $q \times p$ are also whole numbers.
Commutative Property	If p and q are any two whole numbers, then $p + q = q + p$ and $p \times q = q \times p$.
Associative Property	If p , q and r are any two whole numbers, then $(p + q) + r = p + (q + r)$ and $(p \times q) \times r = p \times (q \times r)$.
Distributive property	If p , q and r are any two whole numbers, then $p(q + r) = p \times q + p \times r$.
Additive Identity	If a is any whole number, then $p + 0 = p = 0 + p$.
Multiplicative Identity	If p is any whole number, then $p \times 1 = p = 1 \times p$.
Multiplication by zero	If a is any whole number, then $p \times 0 = 0 = 0 \times p$.
Division by zero	If p is any whole number, then $p \div 0$ is not defined.

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3.

(a) 18

(b) Addition and Multiplication.

(c) 0.

(d) 1.

(e) 0, 0, Identity element for Addition.

5) 75

6)

(a) $168 \times 102 = 168 \times (100 + 2) = 16800 + 336 = 17136$

(b) $625 \times 279 - 625 \times 79 = 625 \times (279 - 79) = 625 \times 200 = 125000$

7)

(i) 999 : 998 & 1000

(ii) 21998 : 21997 & 21999

iii) 4001 : 4000 & 4002

(iv) 500014 : 500013 & 500015

v) 11111 : 11110 & 11112

8) Marks obtained by Seeta: Math = 99

English = 69

Science = 91

$\therefore$ Total marks obtained by Seeta = $99 + 69 + 91 = 259$

Marks obtained by Reema:

Math = 92

English = 32

Science = 84

$\therefore$ Total marks obtained by Reema = $92 + 32 + 84 = 209$

9) 1800 chocolates

10) Amount paid for lunch = 40

Amount paid for dinner = 55

Number of days = 10

Money paid by him in 10 days = $10 \times (55 + 40) = 950$

11) 166,466

12)

a. 20

b. $768 \times 10 = 7680$

c. 2

d. 99,999

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Novice level

Multiple choice type

1	2	3	4	5	6
a	c	c	b	d	b
7	8	9	10		
b	d	c	b		

Fill in the blanks:

(a) 24 (b) Addition, Multiplication

(c) 0 (d) 1

(e) 0, 0, Additive Identity

Subjective Questions:

14	15.a	15.b	17	18	19	20
20	86314	9162500	139, 148	730800	135	665
21	22	25.a	25.b	25.c	28	29
100	20	93155	898200	26980	2345	234
30						
101,0						



Advanced level

1. (i) 500 (ii) 3100 (iii) 510

2. -1

3. $39480 - 38490 = 990$

5. (i) 999900

(ii) 999990

6. 30,00,000

7. (i) 5,423 (ii) 24,57,900 (iii) 1,56,800 (iv) 56,25,000

9. (i) 1,49,700 (ii) 35,800 (iii) 50,00,000

10. 8,91,378

12. 67,530

13. 1,66,666

14. 46,120

15. -1

16. 6,4,2

17. 9879001155

19. 969,987

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20. 119,580

21. 64,396

22. 203

23. 15075 kg

24. 469,102

25. 14,400 g

END